CIVITAS
Sustainable and smart mobility for all
2030

CIVITAS indicators

Theoretical monthly public transport expenditure over average household income (SOC_EQ_AF2)

DOMAIN

Transport	Environment	Energy	**Society**	Economy

TOPIC	**Equity**	
IMPACT	**Affordability of transport services** *Improving the affordability of transport services*	**SOC_EQ**

Category

Key indicator	Supplementary indicator	State indicator

CONTEXT AND RELEVANCE

Affordability of public transport measures how manageable public transport fares are for users relative to their income. Ensuring that fares are affordable allows residents to access essential services, employment opportunities, and social activities without experiencing financial strain. This helps to prevent the social and economic exclusion of low-income households. As such, this indicator evaluates the inclusivity and equity of urban transport systems.

This indicator provides a measure of the cost of transport services. **It is a relevant indicator when the policy action is aimed at reducing or removing economic barriers preventing some groups of citizens to use transport services. A successful action is reflected in a LOWER value of the indicator.**

DESCRIPTION

This indicator is a **dimensionless index** obtained from an estimated **average cost for using public transport services** and the **average income of poorest quintile of the population**.

METHOD OF CALCULATION AND INPUTS

The indicator is calculated as the ratio between an average yearly cost for using public transport and the income of the poorest quintile of population. **The calculation of the average yearly cost and of the ratio is made within the supporting tool building on a set of inputs**.

There are two alternative methods of calculation available for this indicator. The two methods distinguish for the complexity of the quantification and for their significance.

METHOD 1	METHOD 2
Average yearly cost of public transport based on pre-defined shares of ticket types	**Average yearly cost of public transport based on observed shares of ticket types**
Some inputs are theoretical values not tailored to the specific experiment site.	All inputs are tailored to the specific experiment site.
Complexity	Complexity
Significance	Significance

Method 1

Calculation of the index based on Public Transport fares and income statistics using pre-defined ticket shares	Significance: **0.50**

METHOD OF CALCULATION

The indicator is computed according to the following steps (managed in the supporting tool):

- **Quantification of the average yearly cost for using public transport**
- **Estimation of the cost/income ratio**.

INPUTS

The following information should be coded in the supporting tool to compute the indicator:

a) $TckPrice$. **Representative cost of a public transport ticket**. Ticket fares are readily available from transport operators. This element should be the base tariff for one single urban trip.

b) $MonthPass$. **Representative cost of a pass allowing for using urban public transport for one month**. Ticket fares are readily available from transport operators. If only one urban monthly pass exists, then the input is its price. If alternative passes exist, an "average" representative cost should be defined.

c) $MonthPassStd$. **Representative cost of a pass allowing students for using urban public transport for one month**. In many cases, students are entitled to use public transport paying a reduced fare. If discounted passes for students do not exist in the experiment area, this input should be considered equal to $MonthPass$.

d) $MonthPassEld$. **Representative cost of a pass allowing elderly individuals for using urban public transport for one month**. In many cases, elderly individuals are entitled to use public transport paying a reduced fare. If this does not apply in the experiment area, this input should be considered equal to $MonthPass$.

e) $Income$. **Yearly household income of the fifth (i.e., poorest) quintile of population**. Ideally, this should be the income in the experiment area, but if local data does not exist, regional or even national data can be used. National income by quintile in all EU countries can be found in the Eurostat database (**table ilc_di01**):

The experiment would be reflected in the indicator by changing the price of one or more ticket types.

EQUATIONS

The equations **used within the supporting tool** to manage the calculation, building on the provision of the inputs, are the followings:

Estimation of the average yearly cost for using public transport:

$$AvYearPTCost = (TckPrice * TckShr) * 100 + (MonthPass * MonthPassShr + MonthPassStd * MonthPassStdShr + MonthPassEld * MonthPassEldShr) * 12$$

The elements $TckShr, MonthPassShr, MonthPassStdShr$, $MonthPassEldShr$ are weights pre-defined in the tool.

100 reflects the exogenous assumption that single tickets are used for 100 trips per year (2 trips per week) as those travelling by public transport for daily mobility are expected to use monthly or yearly passes.

12 reflects the exogenous assumption that yearly passes are purchased in any month of the year.

Estimation of the cost/income ratio (indicator value):

$$\boldsymbol{PTAff = \frac{AvYearPTCost}{AvIncome} * 100}$$

Method 2

Calculation of the index based on Public Transport fares and income statistics using tailored ticket shares	Significance: **0.75**

METHOD OF CALCULATION

The indicator is computed according to the following steps (managed in the supporting tool):

- **Quantification of the average yearly cost for using public transport**
- **Estimation of the cost/income ratio**.

INPUTS

The following information should be coded in the supporting tool to compute the indicator:

f) $TckPrice$. **Representative cost of a public transport ticket**. Ticket fares are readily available from transport operators. This element should be the base tariff for one single urban trip.

g) $MonthPass$. **Representative cost of a pass allowing for using urban public transport for one month**. Ticket fares are readily available from transport operators. If only one urban monthly pass exists, then the input is its price. If alternative passes exist, an "average" representative cost should be defined.

h) $MonthPassStd$. **Representative cost of a pass allowing students for using urban public transport for one month**. In many cases, students are entitled to use public transport paying a reduced fare. If discounted passes for students do not exist in the experiment area, this input should be considered equal to $MonthPass$.

i) $MonthPassEld$. **Representative cost of a pass allowing elderly individuals for using urban public transport for one month**. In many cases, elderly individuals are entitled to use public transport paying a reduced fare. If this does not apply in the experiment area, this input should be considered equal to $MonthPass$.

j) $TckShr$. **Relative importance of base single trips ticket** for the calculation of average cost. This share should be defined considering the popularity of this type of ticket based on statistics of tickets sold or other information.

k) $MonthPassShr$. **Relative importance of monthly pass** for the calculation of average cost. This share should be defined considering the popularity of this type of pass based on statistics of tickets sold or other information.

l) $MonthPassStdShr$. **Relative importance of student monthly pass** for the calculation of average cost. This share should be defined considering the popularity of this type of pass based on statistics of tickets sold or other information. If a discounted pass for students does not exist in the experiment area this share should be set to zero.

m) $MonthPassShr$. **Relative importance of elderly monthly pass** for the calculation of average cost. This share should be defined considering the popularity of this type of pass based on statistics of tickets sold or other information. If a discounted pass for elderly individuals does not exist in the experiment area this share should be set to zero.

n) $Income$. **Yearly household income of the fifth (i.e., poorest) quintile of population**. Ideally, this should be the income in the experiment area, but if local data does not exist, regional or even national data can be used. National income by quintile in all EU countries can be found in the Eurostat database (**table ilc_di01**):

The experiment would be reflected in the indicator by changing the price of one or more ticket types.

EQUATIONS

The equations **used within the supporting tool** to manage the calculation, building on the provision of the inputs, are the followings:

Estimation of the average yearly cost for using public transport:

$$AvYearPTCost = (TckPrice * TckShr) * 100 + (MonthPass * MonthPassShr + MonthPassStd * MonthPassStdShr + MonthPassEld * MonthPassEldShr) * 12$$

Estimation of the cost/income ratio (indicator value):

$$\boldsymbol{PTAff = \frac{AvYearPTCost}{AvIncome} * 100}$$

ALTERNATIVE INDICATORS

An alternative indicator for measuring the transport affordability is **SOC_EQ_AF3**. This alternative indicator uses cost for using shared mobility services in addition to public transport costs. Therefore, it is more complex to implement but more significant. **If in the experiment area there are no shared transport services, the two indicators are equivalent**.

Furthermore, **SOC_EQ_AF1** can be used when pilot measures target affordability beyond public transport and shared mobility, or for smaller-scale local pilots. It relies on a sample survey of residents and users in the pilot area to collect data on household income and transport expenditure. The indicator measures the share of households spending more than 10% of their income on transport, disaggregated by income group.